

 METHODOLOGY		PLG-METH-0005-V2	
Generator Interconnection Power Factor Analysis Methodology		Version:	2
		Effective Date:	5/17/2019
Author(s):	<i>Damien Sommer, P.E.</i>	Prepared Date:	5/13/2019
Manager(s):	<i>Paul Walter, P.E. Paul F. Walter P.E.</i>	Review Date:	5/17/2019
VP(s):	<i>Andy Dolan Andy Dolan</i>	Approval Date:	5/17/2019

Table of Contents

1. Purpose.....	3
2. Scope.....	3
3. Applicability	3
3.1. Regulatory Applicability	3
3.2. Affected Assets	3
3.3. Affected Personnel	3
4. Criteria	3
5. Methodology – PF Criteria Evaluation	4
5.1. Modeling and Case Preparation	4
5.2. Performing PF Criteria Evaluation at Max Real Power Output.....	6
5.3. Performing PF Criteria Evaluation at Min Real Power Output.....	8
6. Methodology – Stability Assessment	8
7. Capacitor Location.....	9
8. Operation Concerns	9
9. Administration	9
9.1. Review	9
9.2. Retention	10
10. Approvals/Revision History	10
11. Example A – Generator Meets PF criteria	11
12. Example B – Generator Requires Static Resources	13
13. Example C – Generator Requires Dynamic Resources	15

14. Example D – Generator Requires Static and Dynamic Resources	17
15. Example E – Test for Inductive Output of Asynchronous Machines at Low Levels of Output	19

1. Purpose

The purpose of this methodology is to define the process ATC uses to determine that generators meet ATC and NERC power factor (PF) criteria. It also defines how ATC calculates the required additional dynamic and static reactive sources that are required if a generator is found to not meet the ATC and NERC PF requirement.

2. Scope

This methodology only applies to generation facilities that are interconnected to the ATC system and includes both synchronous and asynchronous generating facilities.

The methodology covers generation interconnection power factor analysis, and does not directly apply to interconnection point voltage analysis.

3. Applicability

3.1. Regulatory Applicability

NERC Reliability Standard FAC-002-2

3.2. Affected Assets

Generation facilities that are connected to the ATC system and the transmission facilities that may be affected by the generation interconnection

3.3. Affected Personnel

ATC System Planning staff

4. Criteria

From the ATC Planning Criteria Version 19.2:

Power Factor Requirements for Interconnection Generating Units are as follows. ATC's standard power factor range for synchronous and non-synchronous (e.g., wind turbines, solar) generation is 0.95 leading (when a Generating Facility is consuming reactive power from the Transmission System) to 0.95 lagging (when a Generating Facility is supplying reactive power to the Transmission System).

Static reactive power sources can only be used to make up for losses between the machine terminal and the POI for synchronous machines and losses between the terminal of the machines and the high side of the GSU for non-synchronous machines. All other reactive power to meet the power factor requirement must be provided by continuous and sustainable dynamic sources. Operation across the entire power factor range must be fully dynamic, variable, and capable of sustained indefinite operation.

Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Cap Banks are considered static sources of reactive power.

Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters, and synchronous condensers are all considered dynamic sources of reactive power.

The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing continuous dynamic reactive power at the following locations:

- a) The Point Of Interconnection (POI) for all synchronous generators

For synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI.

- b) The high-side of the generator step up transformer (GSU) for all nonsynchronous generators

For non-synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the GSU.

Dynamic reactive power provided by non-synchronous generators must meet the following requirement from FERC order 827 Item 35:

"Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses."

5. Methodology – PF Criteria Evaluation

5.1. Modeling and Case Preparation

Perform the following steps to prepare cases that will be used to determine if the PF requirement is met:

1. Create a simplified PSSE .sav case modeling the generator and all known facilities from the generator terminal to the Point of Interconnection (POI). Represent the rest of the system with an infinite machine model placed on the POI bus. The following record shows an example of a infinite machine model for PSSE version 33:

```
990000,'1 ', 0, 0, 9999.000, -9999.000,0.95, 991000, 100.0, 0, 1.0, 0.0, 0.0,1.00000,1,
100.0, 9999.000, -9999.000, 1,1.0000
```

When modeling the generator under study, rounding will be used to determine the machines power factor range to two decimal places. For example, 0.951 will be rounded to 0.95 and 0.955 will be rounded to 0.96.

If voltage dependent reactive capability curves (P-Q curves) are given for the machine the curve for 1.0 pu voltage should be used for determining reactive limits of the machine unless more conservative limits are stipulated by the developer. or other machine or plant limits provided by the developer should be used to determine the reactive capability to include in the steady state model of the machine.

2. Test for capacitive output of all machines
 - a. Set the generation request machines to 100% MW output
 - b. Set the infinite machine to control voltage on the bus where the PF is to be calculated from to 0.95 p.u. Set all the dynamic and static reactive devices provided by the generator owner to maximum MVAR output by setting QGEN and QMIN equal to QMAX and solve the PSSE case until it comes to solution in 1 iteration. Save a copy of this .sav case with “_QMAX” at the end of the file name.
3. Test for inductive output of all machines
 - a. Set the generation request machines to 100% MW output
 - b. Set the infinite machine to control voltage on the bus where the PF is to be calculated to 1.05 p.u. Set all the dynamic and static reactive devices provided by the generator owner to minimum MVAR output by setting QGEN and QMAX equal to QMIN and solve the PSSE case until it comes to solution in 1 iteration. Save a copy of this .sav case with “_QMIN_PMAX” at the end of the file name.
4. Test for inductive output of asynchronous machines at low levels of output
 - a. Review the P-Q capability curve for the machine. If the P-Q Curve is rectangular or D-shaped this analysis will not be required. If the P-Q Curve is V-shaped or U-shaped this analysis will be required. See figure 1 below for examples of the different P-Q curve types.

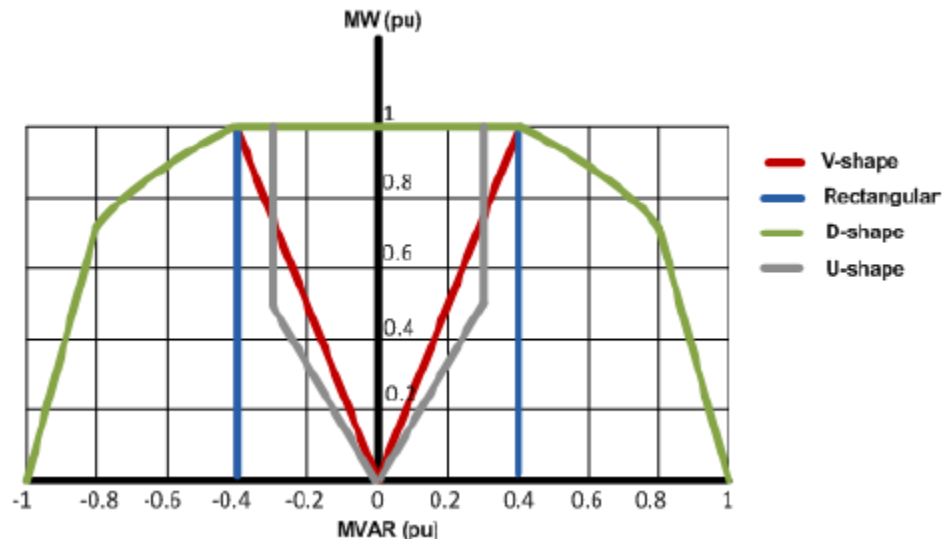


Figure 1, Example P-Q Curve Types for Asynchronous Machines

- b. If the analysis is required, set the generation request machines to 0 MW output.

- c. Set the infinite machine to control voltage on the bus where PF is to be calculated to 1.05 p.u. Set all dynamic and static reactive devices provided by the generator owner to minimum MVAR output and solve the PSSE case until it comes to solution in 1 iteration. Save a copy of this .sav case with “_QMIN_PMIN” at the end of the file name.
5. Make a slider diagram showing all the facilities in the .sav case. Save the slider diagram.

5.2. Performing PF Criteria Evaluation at Max Real Power Output

Perform the following calculations to determine if the generation request meets the PF requirement at 100% MW output or, if it doesn't, how much additional static and dynamic resources are required:

1. Determine the real and reactive power at the bus where PF criteria is to be applied. This will be at the POI for synchronous machines or at the high side of the generator step-up (GSU) transformer for asynchronous machines. This is observed from the PSSE simulation.
 - a. Real power
 - b. Reactive power
2. Calculate how many MVARs are needed at the measured bus to meet the PF requirement given the MWs that reach the bus. This is the required amount of dynamic MVARs. The required MVARs are calculated as follows:

$$\text{Required MVA}_{\text{capacitive}} = \text{MW}/0.95$$

$$\text{Required MVA}_{\text{inductive}} = \text{MW}/0.95$$

$$\text{Required Dynamic MVAR} = \sqrt{\text{MVA}^2 - \text{MW}^2}$$

$$\text{Or by } Q = \text{MW} \times \tan(\cos^{-1}(\text{PF}))$$

3. Determine if the generator facilities can produce enough dynamic MVARs to meet the capacitive PF requirement and absorb enough dynamic MVARs to meet the inductive PF requirement.
 - a. Sum up the maximum nominal dynamic MVAR output from all sources provided by the generation request (at the source terminals in the PSSE simulation). This sum should not include any auxiliary loads. If this sum is greater than or equal to the required MVARs calculated in step 2 then the generation request can produce enough dynamic MVARs to meet the capacitive PF requirement.
 - b. Sum up the minimum nominal dynamic MVAR output from all sources provided by the generation request (at the source terminals in the PSSE simulation). This sum should not include any auxiliary loads. If this sum is less than or equal to the required

MVARs calculated in step 2 then the generation request can absorb enough dynamic MVARs to meet the inductive PF requirement.

If both requirements are met skip to step 5. If one or both of the requirements are not met, then proceed to step 4.

4. Determine the minimum additional dynamic reactive resources that would be required to meet ATC PF criteria.

If the generation request is deficient in dynamic resources the minimum additional dynamic reactive resources required are calculated as follows:

$$\text{Additional Required Dynamic MVARs} = \text{Required Dynamic MVARs} - (\text{MVAR}_1 + \text{MVAR}_2 + \dots + \text{MVAR}_n)$$

Where Required Dynamic MVARs is the value found in item 2 and $(\text{MVAR}_1 + \text{MVAR}_2 + \dots + \text{MVAR}_n)$ is the sum found in either item 3.a or 3.b.

5. Determine if enough dynamic reactive resources are being produced or absorbed at the bus where the criteria apply.
 - a. If enough nominal dynamic MVARs are available as determined in step 3 then simply determine if the MVARs at the criteria bus (from the PSSE simulation) are equal to or greater than required dynamic MVARs. If so, then the generation request meets the ATC PF criteria and no further analysis is required.
 - b. If enough nominal dynamic MVARs are available at the source terminals but there are not enough MVARs at the criteria bus (from the PSSE simulation) then the generation request does not meet the ATC PF criteria because of losses. In that case proceed to step 6.a.
 - c. If not enough nominal dynamic MVARs are available at the source terminals and there are not enough MVARs at the criteria bus (from the PSSE simulation) then the generation request does not meet the ATC PF criteria. In that case proceed to step 6.b.
6. Determine the amount of static resources required to meet ATC PF criteria.
 - a. If the generation request has enough nominal dynamic reactive resources then the amount of static reactive resources required is calculated as follows:

$$\text{Required additional Static MVARs} = \text{Required Dynamic MVARs} - \text{Dynamic MVARs}_{\text{Criteria Bus}}$$

Where Required Dynamic MVARs is from item 2 and Dynamic MVARs_{Criteria Bus} is observed from the PSSE simulation (eg. the total Mvars at the criteria bus as observed in the PSSE simulation).

- b. If the generation request does not have enough nominal dynamic resources then the amount of static reactive resources required is calculated as follows:

Required Static MVARs = Required Dynamic MVARs –

(Dynamic MVARs_{Criteria Bus} + Additional Required Dynamic MVARs)

Where Required Dynamic MVARs is from item 2, Additional Required Dynamic MVARs is from item 4, and Dynamic MVARs_{Criteria Bus} is observed from the PSSE simulation (eg. the total Mvars at the criteria bus as observed in the PSSE simulation).

7. Determine the bus where the static MVARs will be connected. Record the per unit bus voltage where the additional static resources are expected to be located in per unit.
8. Rating of additional static Mvars required to make up for reactive plant losses given at nominal voltage (1.0 pu). Calculated as: $Mvar_{Nom} = \frac{[total\ from\ item\ 6]}{([value\ of\ item\ 7])^2}$
9. The total value of reactive compensation for losses, given at nominal voltage, is the sum of the static devices modeled in the base case plus the additional value calculated in item 8.

5.3. Performing PF Criteria Evaluation at Min Real Power Output

If PF analysis is required for low MW output levels of asynchronous machines (eg. section 5.1 item 4) perform the following calculations to determine if the generation request meets the PF requirement or, if it doesn't, how much additional static and dynamic resources are required:

1. Determine if MVARs are produced by the plant facilities at the bus where the PF criteria is to be applied. If 0 MVARs are produced no static resources are required and the analysis is complete. Otherwise, continue to step 2.
2. If MVARs are contributed at the bus where PF criteria is to be applied this is the amount of static MVAR consumption will be required of the interconnection customer.

6. Methodology – Stability Assessment

When cap banks are used by a generating facility to meet the PF criteria the power plant cap banks will be dispatched at the most conservative scenario in the stability assessment. Unless the study engineer determines that a more conservative scenario exists because of prior

operating experience or engineering judgement ATC will follow these procedures to simulate the most conservative dispatch of power plant cap banks:

Synchronous machines are less stable when absorbing MVARs. Therefore, the more conservative scenario would include power plant cap banks switched on which would lower the MVAR output of the synchronous machines studied. ATC will study synchronous machines in stability assessments with all power plant cap banks switched on to maximum MVAR output.

Asynchronous machines are considered less stable when power plant voltages are lower because there is greater risk that the machine will trip because of low voltage relays. Therefore, the more conservative scenario would include power plant cap banks switched off which would result in lower power plant voltages. ATC will study asynchronous machines in stability assessments with all power plant cap banks switched off.

7. Capacitor Location

For new generator interconnections, ATC has consistently required separate ATC and Customer substations. Section 3.3 of the ATC Generating Facility Interconnection Guide states that “ATC’s interconnection substation will be designed as an entirely separate substation from the Customer’s substation.” ATC does not desire to have Customer-owned equipment in the ATC substation.

8. Operation Concerns

The PF requirement is intended to make sure that power plants have enough reactive capability in reserve to respond to steady state and dynamic stability events. Therefore, ATC may require the Generator Owner to have an operating procedure for switching in cap banks that ensures adequate dynamic reactive reserves. These operating procedures should ensure:

1. Cap banks are switched in before the dynamic reactive resources are dispatched to maximum MVAR output.

Cap banks are not switched in such that the dynamic reactive resources begin to consume MVARs.

The developers should design the voltage controller such that the switched capacitors do not conflict with how the generator controls voltage such that ATC will see the full power factor range available at all of the ranges of system loads and system conditions. When switching capacitors IN or OUT, developers should make sure that they are meeting the full power factor range specified in GIA, FERC Order 827, ATC Planning Criteria, and their obligations under NERC VAR-002-4.

9. Administration

9.1. Review

Review at least every five years.

9.2. Retention

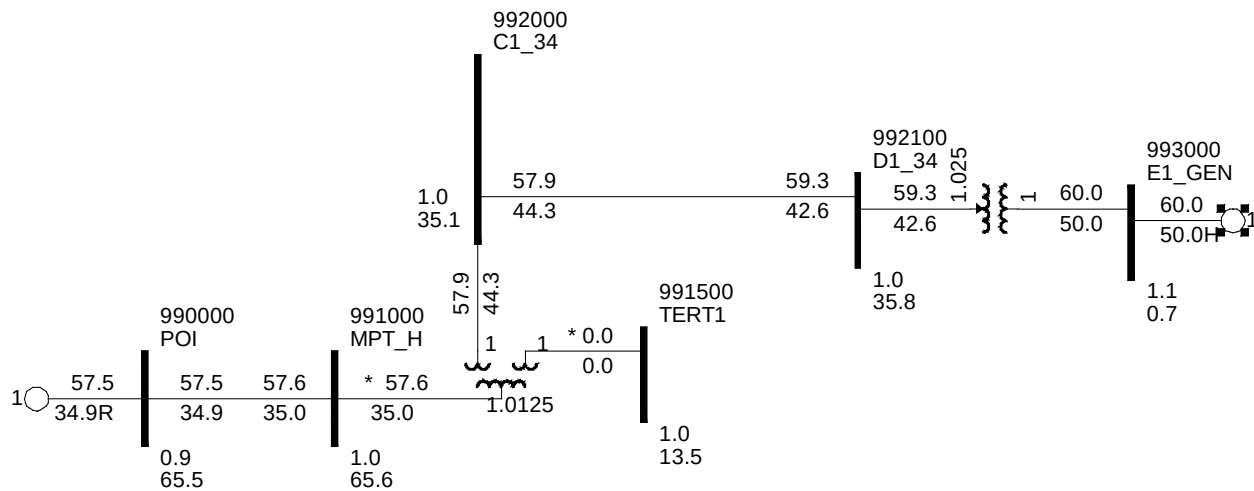
The previous version of this document will be retained until this version is superseded.

10. Approvals/Revision History

Revision	Author(s)	Reviewer(s)/ Approver(s)	Effective Date	Summary of Changes
1	Damien Sommer, Adam Manty, Jamal Khudai, Ruchi Singh, Mike Marz	Paul Walter, Andy Dolan	3/20/2018	Initial document
2	Damien Sommer	Paul Walter, Andy Dolan	5/15/2019	Updated to reflect ATC Power Factor Criteria change to +/- 0.95 implemented in PLG-CR-0001- V19.2

11. Example A – Generator Meets PF criteria

An asynchronous wind farm has requested to connect to POI bus 990000. The generation request must meet the ATC PF requirement at the high side of the GSU at bus 991000. The generation request has a nominal dynamic MVAR capability of 50.00 MVARs and a nominal static MVAR capability of 0.00 MVARs.



1. The power at the high side of the GSU is 57.6 MW.
2. Required MVA $57.6\text{MW}/0.95 = 60.63\text{ MVA}$

$$\text{Required MVAR at high side of GSU} = \sqrt{60.63^2 - 57.6^2} = 18.93\text{ MVAR}$$

3. 18.93 Dynamic MVAR Required < 50.00 Nominal Dynamic MVAR Resources Provided

Therefore, more dynamic MVARs are not required.

4. Not applicable

5. 18.93 Dynamic MVAR Required < 35.00 Observed Dynamic MVAR

Therefore, more static MVARs are not required.

6. Not applicable

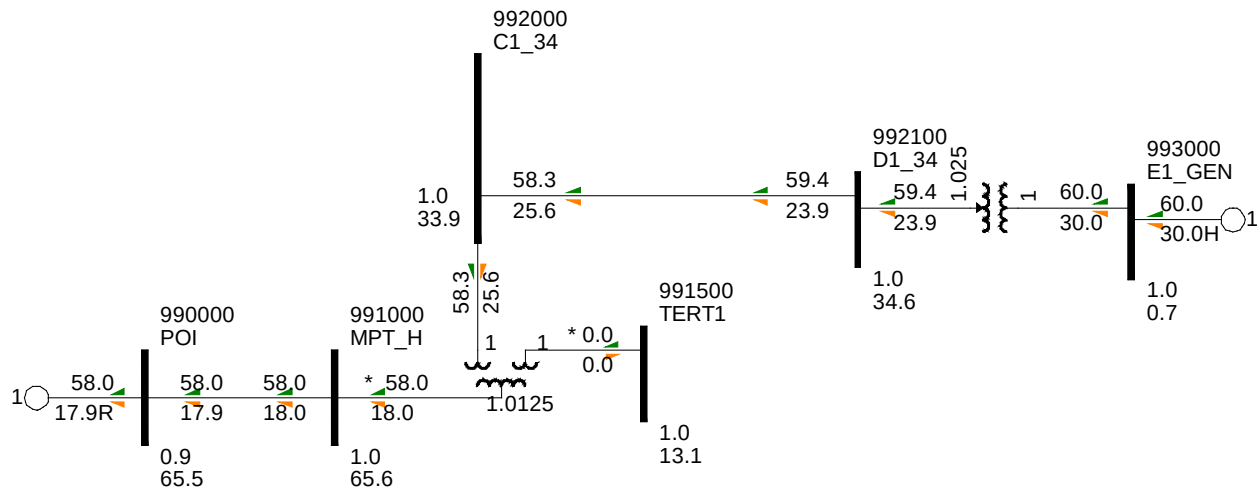
7. Not applicable

8. Not applicable

9. Not applicable

12. Example B – Generator Requires Static Resources

An asynchronous wind farm has requested to connect to POI bus 990000. The generation request must meet the ATC PF requirement at the high side of the GSU at bus 991000. The generation request has a nominal dynamic MVAR capability of 30.00 MVARs and a nominal static MVAR capability of 0.00 MVARs.



1. The power at the high side of the GSU is 58.0 MW.
2. Required MVA $58.0/0.95 = 61.05$ MVA

$$\text{Required MVAR at high side of GSU} = \sqrt{61.05^2 - 58.0^2} = 19.06 \text{ MVAR}$$

3. 19.06 Dynamic MVAR Required < 30.00 Nominal Dynamic MVAR Resources Provided

Therefore, more dynamic MVARs are not required.

4. Not applicable
5. 19.06 Dynamic MVAR Required > 18.0 Observed Dynamic MVAR

Therefore, more static MVARs are required.

6. Additional Required Static MVARs at high side of GSU =

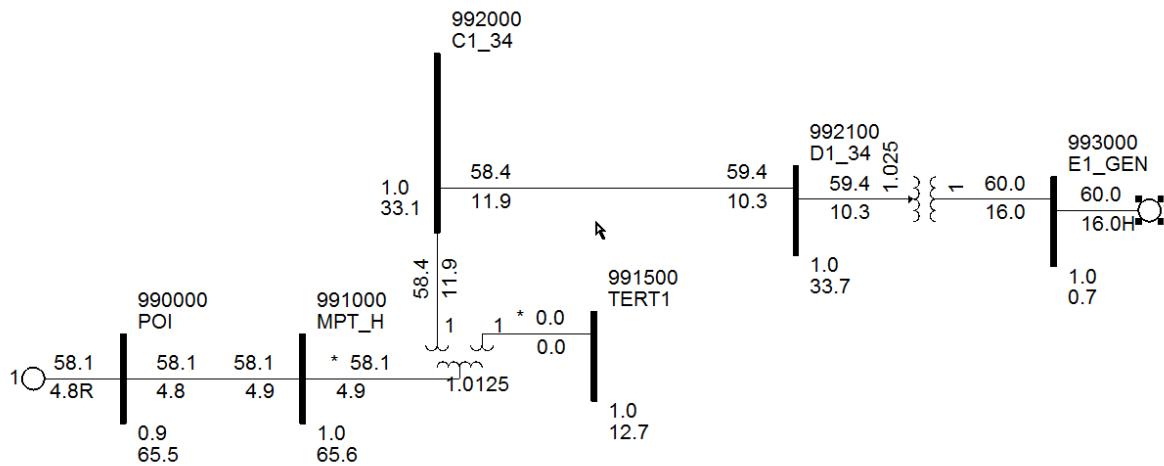
$$19.06 \text{ MVARs} - 18.0 \text{ MVARs} = 1.06 \text{ MVARs}$$

7. Assuming the static cap banks will be installed on the tertiary bus, the per unit voltage is 0.99.
8. Additional Required Static MVARs at Tertiary Bus =
$$1.06 / (0.99)^2 = 1.08 \text{ MVARs}$$
9. 1.08 MVARs of static resources are required.

9. Not applicable

14. Example D – Generator Requires Static and Dynamic Resources

An asynchronous wind farm has requested to connect to POI bus 990000. The generation request must meet the ATC PF requirement at the high side of the GSU at bus 991000. The generation request has a nominal dynamic MVAR capability of 16.00 MVARs and a static MVAR capability of 0.00 MVARs.



1. The power at the high side of the GSU is 58.1 MW.
2. Required MVA $58.1/0.95 = 61.16$ MVA

$$\text{Required MVAR at high side of GSU} = \sqrt{64.56^2 - 58.1^2} = 19.10 \text{ MVAR}$$

3. 19.10 Dynamic MVAR Required > 16.00 Nominal Dynamic MVAR Resources Provided

Therefore, more dynamic MVARs are required.

4. Additional Required Dynamic MVARs = 19.10 MVARs – 16.00 MVARs = 3.10 MVARs
5. 19.10 Dynamic MVAR Required > (4.9 Observed Dynamic MVAR + 3.10 Dynamic MVARs required)

Therefore, more static MVARs are required.

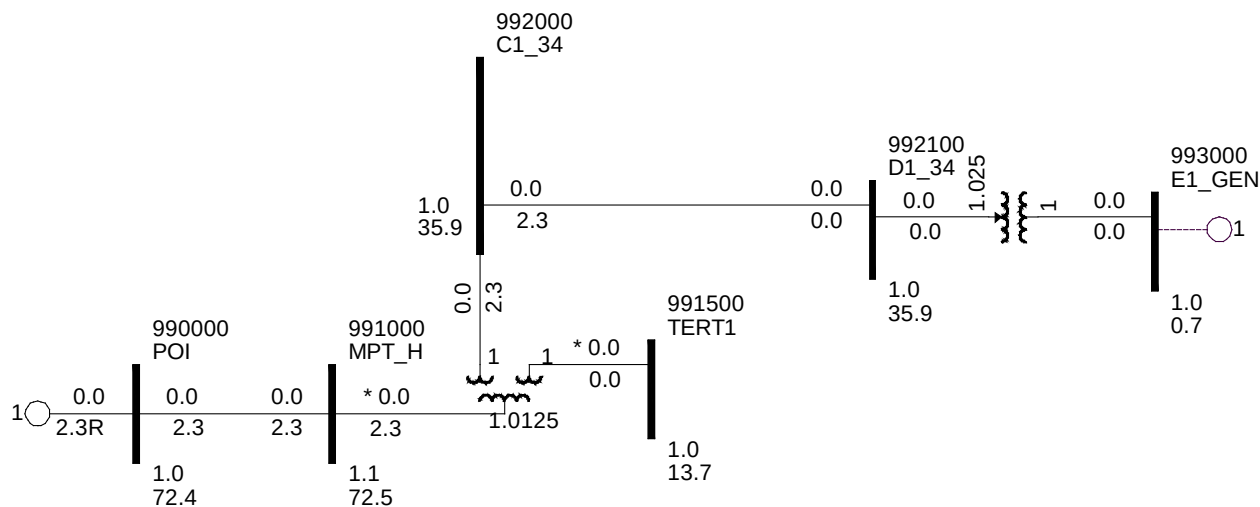
6. Additional Required Static MVARs =

$$19.10 \text{ MVARs} - 4.9 \text{ MVARs} - 3.10 \text{ MVARs} = 11.1 \text{ MVARs}$$

7. Assuming the static cap banks will be installed on the tertiary bus, the per unit voltage 0.97.
8. Additional Required Static MVARs at Tertiary Bus = $11.1 / (0.97)^2 = 11.80$ MVARs
9. 11.80 MVARs required

15. Example E – Test for Inductive Output of Asynchronous Machines at Low Levels of Output

An asynchronous wind farm has requested to connect to POI bus 990000. The generation request must meet the ATC PF requirement at the high side of the GSU at bus 991000 across the full spectrum of MW output. The machine has a V-shaped P-Q curve so the PF criteria needs to be evaluated at near 0 MW output.



1. The G-T request contributes 2.3 MVARs to the system so additional static reactive resources are required to meet the PF criteria.
2. The G-T request will be required to install a 2.3 MVAR inductor.